



THE FLYING PANCAKE

VOUGHT V-173



*DESIGN &
ANALYSIS*

GROUP - 7





MEMBERS OF G-7

Aritra Biswas

22AE10005

**Cheedu Nishkranth
Reddy**

22AE10008

Megavath Sai Ram

22AE10020

Mudavath Swathi

22AE10024

Nagoju Hemanth

22AE10025

Sohan Mondal

22AE10037

Gundla Sanjeev

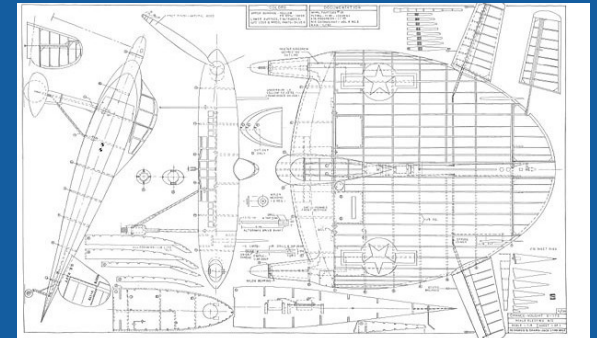
22AE30011

**K. Mohana
Priya Nandini**

22AE30015

BRIEF OVERVIEW OF V-173

- The "Flying Pancake" was an experimental aircraft designed by Charles H. Zimmerman during WW-II for US Navy
- It has a unique, almost circular wing planform with a very low aspect ratio to achieve high lift and short takeoff and landing distances.
- It features fixed, tricycle landing gear which was unusual for the time but provided better ground handling characteristics.
- The V-173 was designed to achieve very low stall speeds, allowing it to take off and land at very low speeds (around 60m / 200ft so close to VTOL). However, its maximum speed and overall performance were limited by the low-power engines used in the prototype.





WEEKLY ROADMAP

01 Define project objectives

02 Study the aerodynamics of Vought-173

03

04

05

06

07

08

09

10

CFD Analysis ,
feedback gain &
design refinement

Building the prototype,
finalising the design and
attempting take-off

GOALS AND OBJECTIVES

Study the design principles and aerodynamic characteristics of V-173

Gather relevant data and prepare the CAD model

Run CFD simulations to understand baseline performance

Optimize the body shape and control surface and gain feedback

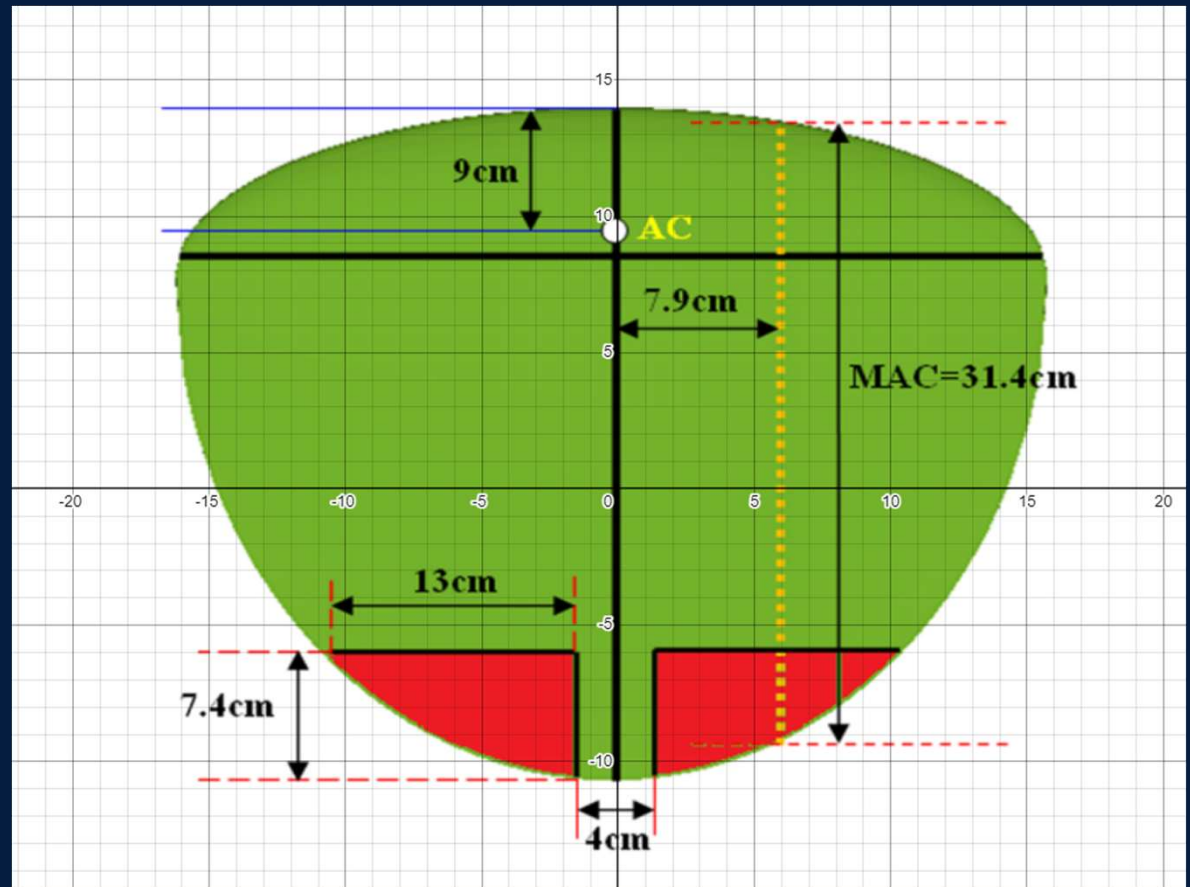
Refine the design to improve VTOL performance and overall speed

Analyze flow patterns , pressure distributions, potential vortices at different AoA

Compile optimized data and make a small prototype for flight testing

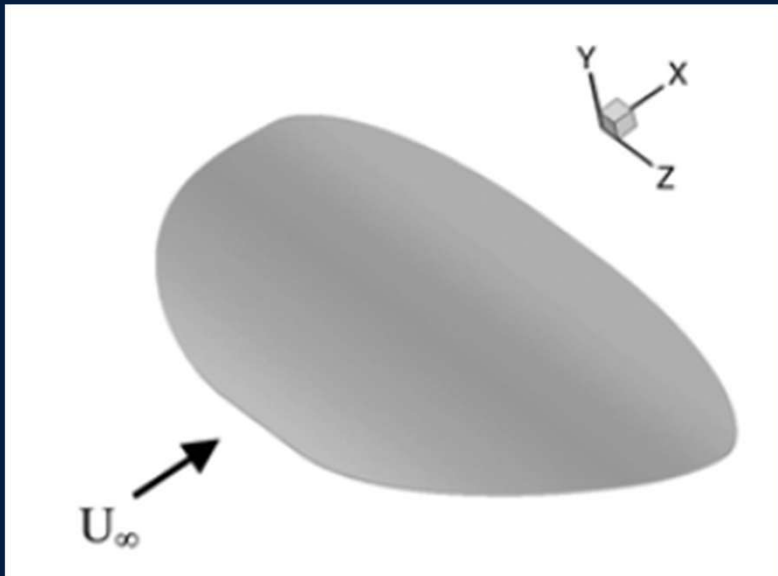
CONTROL SURFACE OF THE MAV DESIGN

The design is
plotted
on Desmos .
It can be found
here : [Link](#)

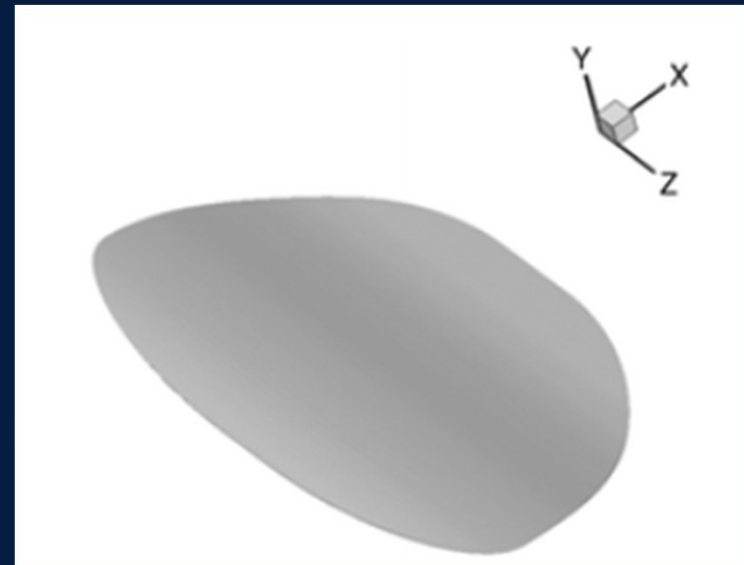


Different Possible Planforms

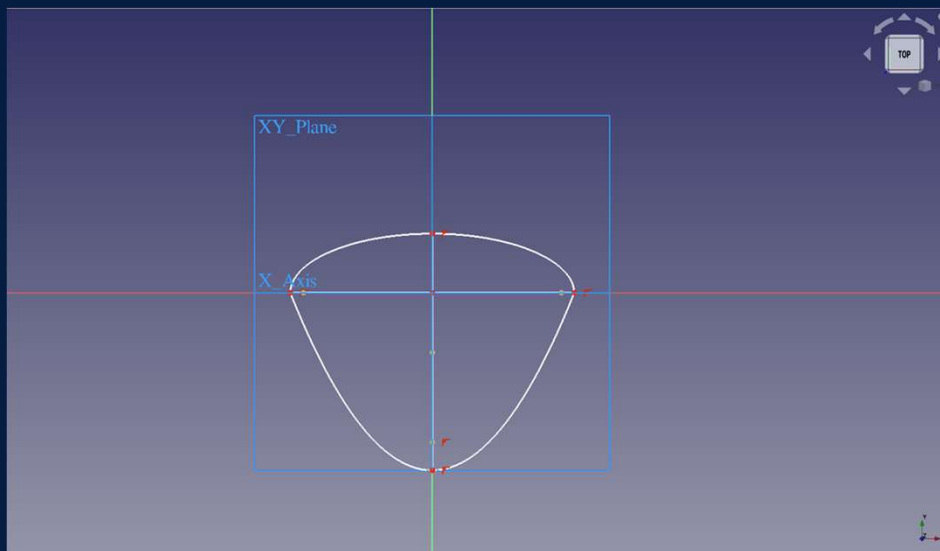
- Inverse Zimmerman



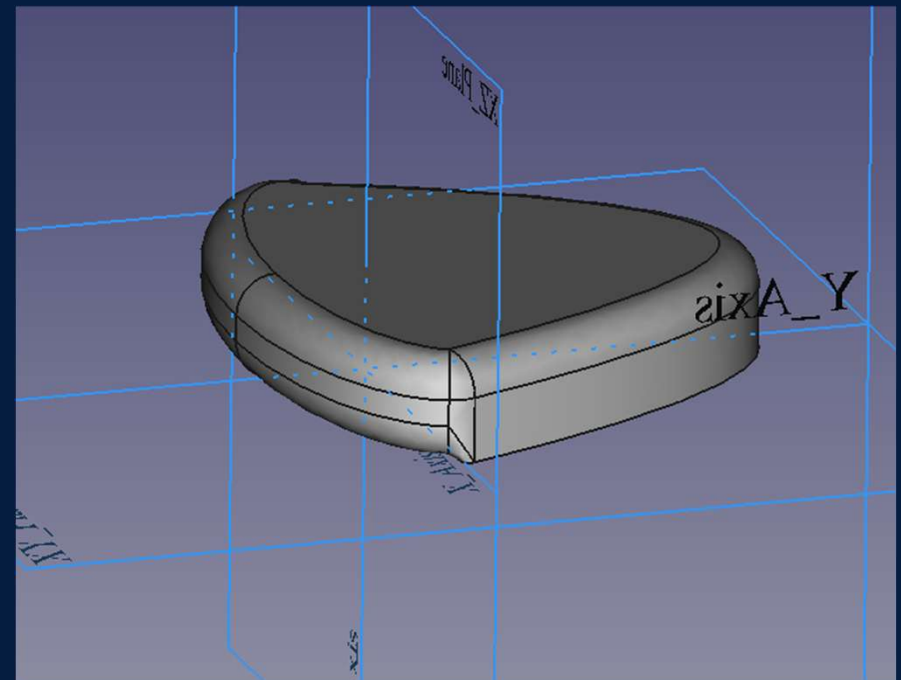
- Zimmerman



CAD MODEL OF THE PLANFORM

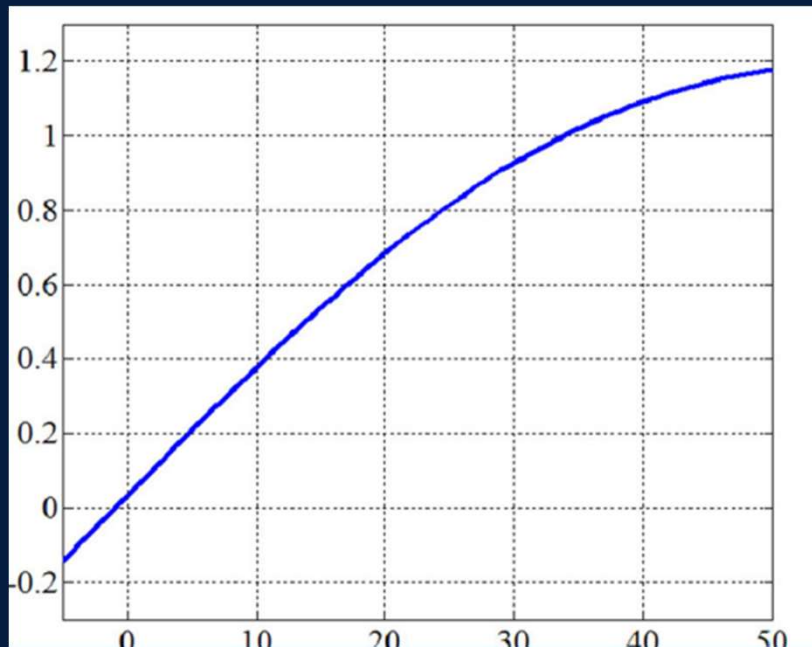


Sketch

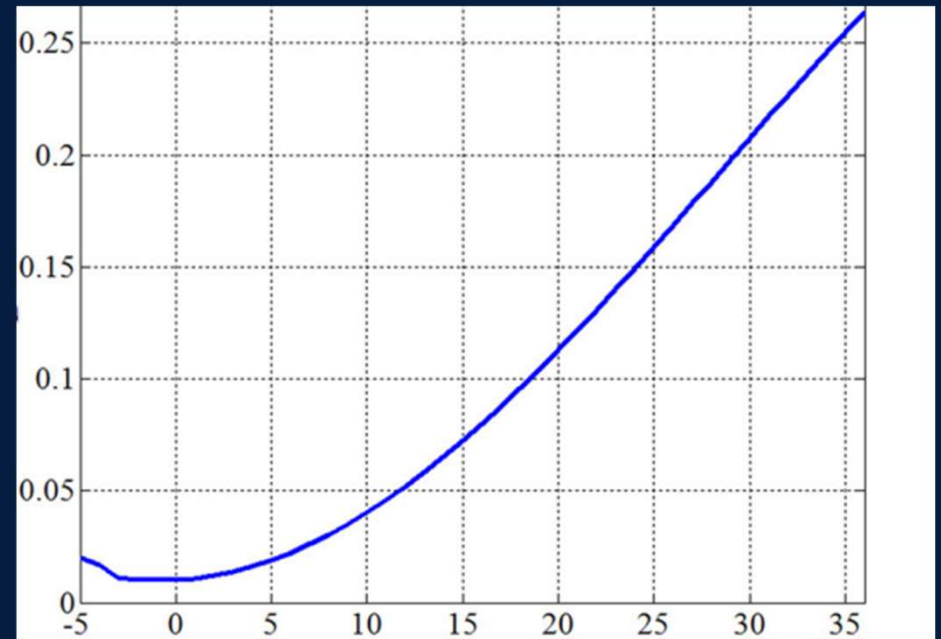


First try of 3D model

PREVIOUSLY FOUND LIFT-DRAG RESULTS



C_L vs AoA



C_d vs AoA

Characteristics for Simulation (ANSYS Fluent)

These are the following aerodynamic characteristics we are going to simulate using ANSYS Fluent :

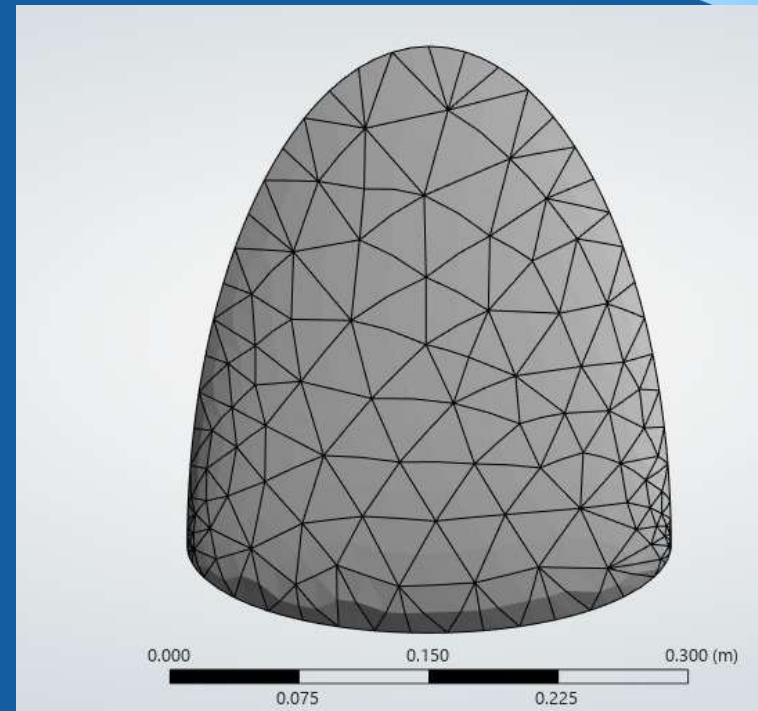
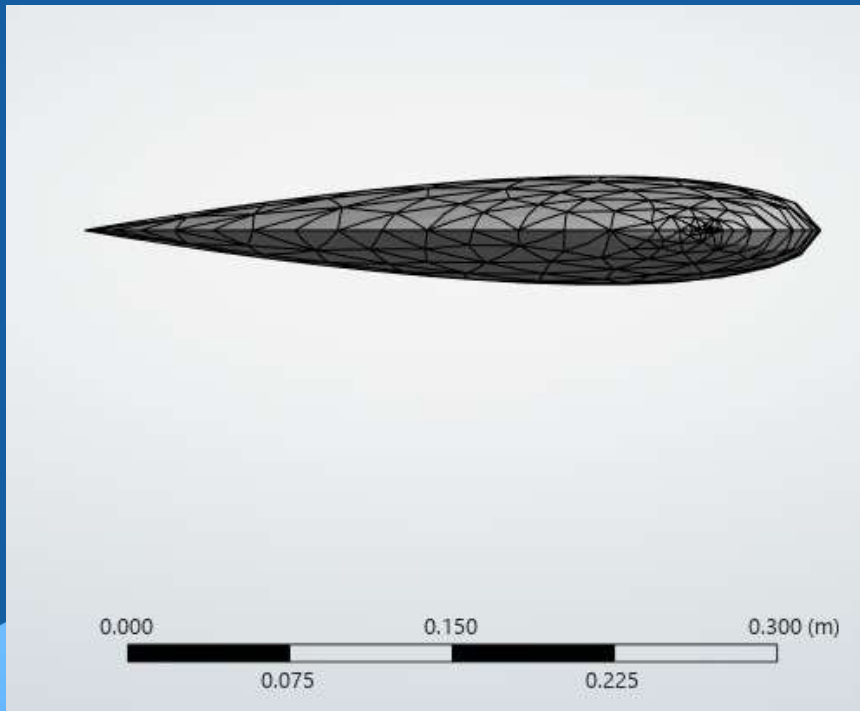
1. Lift and Drag : Determine how efficiently the aircraft generates lift, especially at various angles of attack ; Analyze drag forces, including parasitic, induced, and form drag, to understand how they affect performance.
2. Pressure Distribution : Examine how pressure varies across the wing and fuselage surfaces to identify potential areas of flow separation or high drag; calculate C_p values around the aircraft to understand pressure variations.
3. Flow Visualization : Visualize airflow patterns, especially around the wing edges and wingtips, to study vortex formation and its impact on stability and

control ; investigate boundary layer development, laminar-to-turbulent transition, and possible separation points.

4. Stability and Control : Analyze how the aircraft's pitch changes with different angles of attack, crucial for understanding longitudinal stability ; determine the aerodynamic moments generated about the yaw and roll axes, especially important for assessing the control effectiveness of rudders and ailerons.

5. Performance Metrics : Measure C_l / C_d , study the stall angle, behavior at stall, and post-stall characteristics to ensure safe handling , simulate the takeoff run, climb rate, and landing approach to validate that the model can achieve these maneuvers effectively.

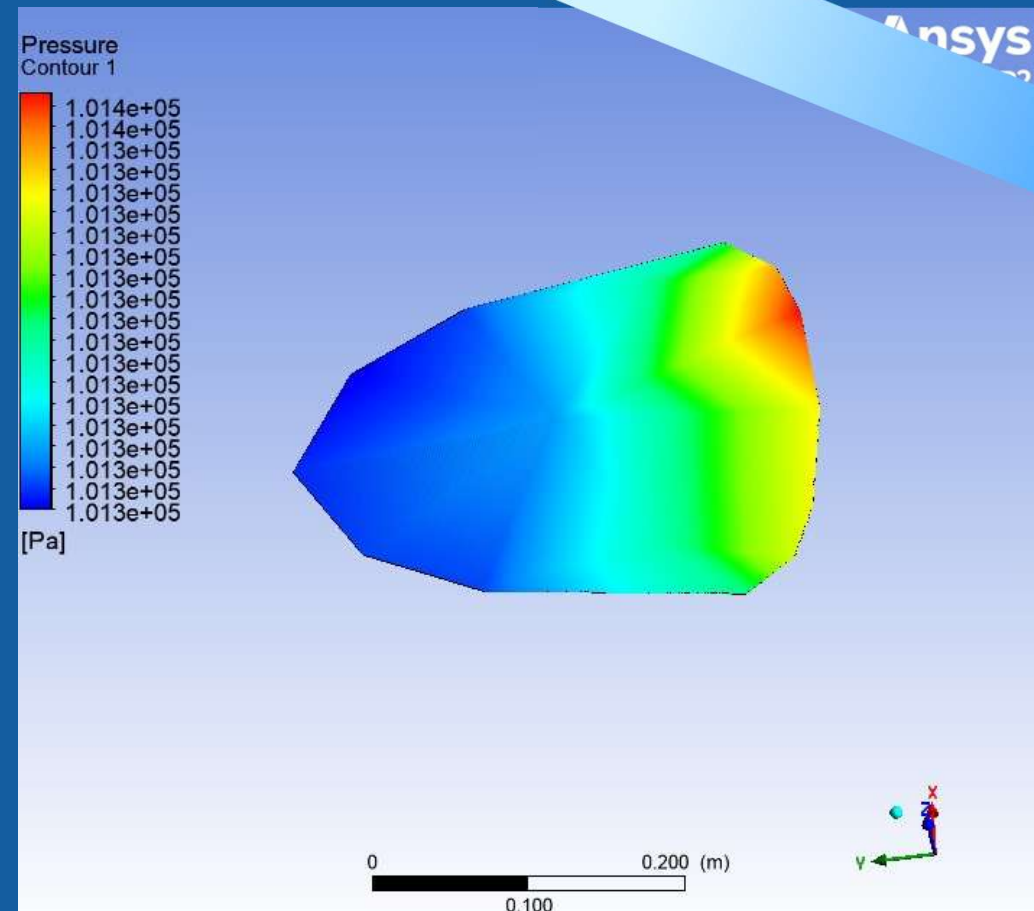
Simulations in ANSYS Fluent



Simulations in ANSYS Fluent

Table 4. Boundary Physics for CFX

Domain	Boundaries	
Default Domain	Boundary - inlet	
	Type	INLET
	Location	Inlet
	Settings	
	Flow Regime	Subsonic
	Mass And Momentum	Normal Speed
	Normal Speed	2.0000e+01 [m s^-1]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio
	Boundary - farfield1	
	Type	OPENING
	Location	farfield1
	Settings	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Subsonic
	Mass And Momentum	Opening Pressure and Direction
	Relative Pressure	1.0000e+00 [atm]
	Turbulence	Medium Intensity and Eddy Viscosity Ratio
	Boundary - farfield2	
	Type	OPENING
	Location	farfield2
	Settings	
	Flow Direction	Normal to Boundary Condition
	Flow Regime	Subsonic
	Mass And Momentum	Opening Pressure and Direction



Simulations in ANSYS Fluent

Boundary - farfield2	
Type	OPENING
Location	farfield2
Settings	
Flow Direction	Normal to Boundary Condition
Flow Regime	Subsonic
Mass And Momentum	Opening Pressure and Direction
Relative Pressure	1.0000e+00 [atm]
Turbulence	Medium Intensity and Eddy Viscosity Ratio
Boundary - outlet	
Type	OUTLET
Location	outlet
Settings	
Flow Regime	Subsonic
Mass And Momentum	Average Static Pressure
Pressure Profile Blend	5.0000e-02
Relative Pressure	1.0000e+00 [atm]
Pressure Averaging	Average Over Whole Outlet
Boundary - Default Domain Default	
Type	WALL
Location	F31.22, F32.22
Settings	
Mass And Momentum	No Slip Wall
Wall Roughness	Smooth Wall

2. Mesh Report

Table 2. Mesh Information for CFX

Domain	Nodes	Elements
Default Domain	508	1751

3. Physics Report

Table 3. Domain Physics for CFX

Domain - Default Domain	
Type	Fluid
Location	B22
Materials	
Air at 25 C	
Fluid Definition	Material Library
Morphology	Continuous Fluid
Settings	
Buoyancy Model	Non Buoyant
Domain Motion	Stationary
Reference Pressure	1.0000e+00 [atm]
Heat Transfer Model	Isothermal
Fluid Temperature	2.5000e+01 [C]
Turbulence Model	SST
Turbulent Wall Functions	Automatic

Pressure
Contour 1

1.014e+05
1.014e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05
1.013e+05

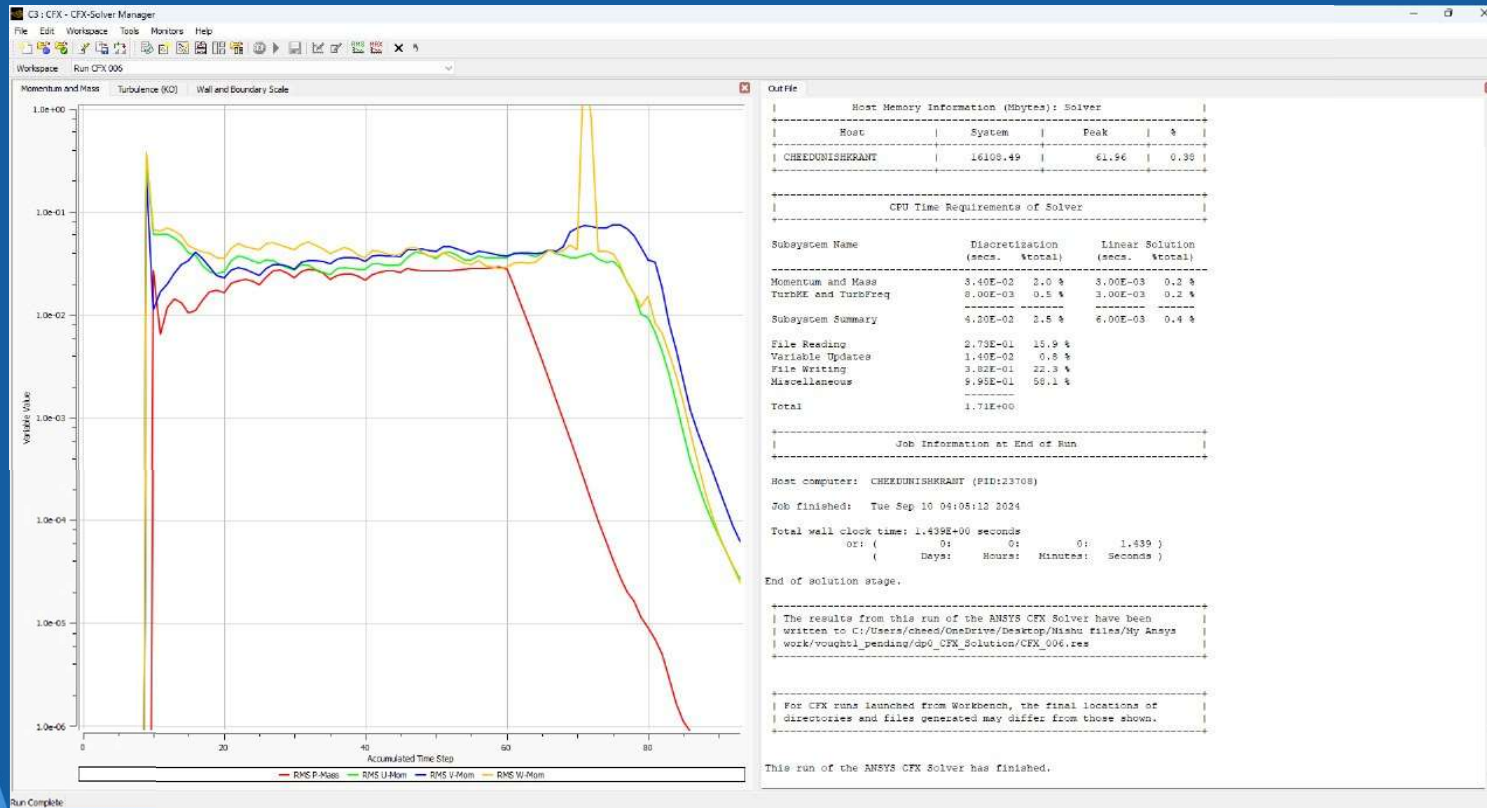
[Pa]

0 0.200 (m)
0.100

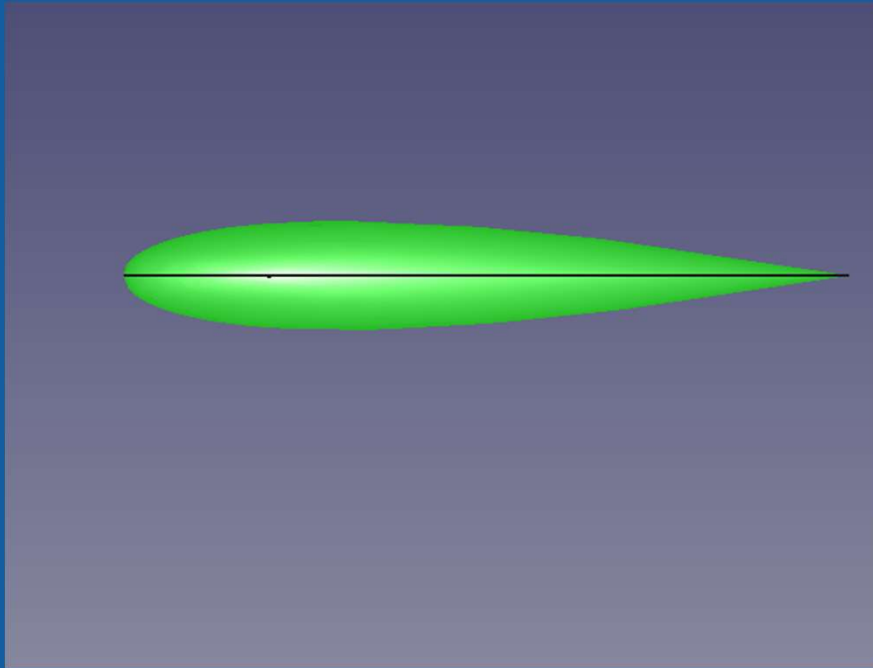
Ansys
2024 R2
STUDENT

The image displays a 3D pressure contour plot of a hexagonal object, likely a component in a fluid flow simulation. The object is rendered with a color gradient representing pressure values, ranging from blue (lower pressure) to red (higher pressure). The highest pressure is concentrated on the top surface of the hexagonal prism, while the lowest pressure is on the bottom surface. The plot is set against a light blue background. A color bar on the left indicates the pressure scale in Pascals (Pa), with values ranging from 1.013e+05 to 1.014e+05. A scale bar at the bottom indicates a length of 0.200 meters, with a midpoint at 0.100 meters. The ANSYS logo and version information (2024 R2) are visible in the top right corner, along with the text 'STUDENT'.

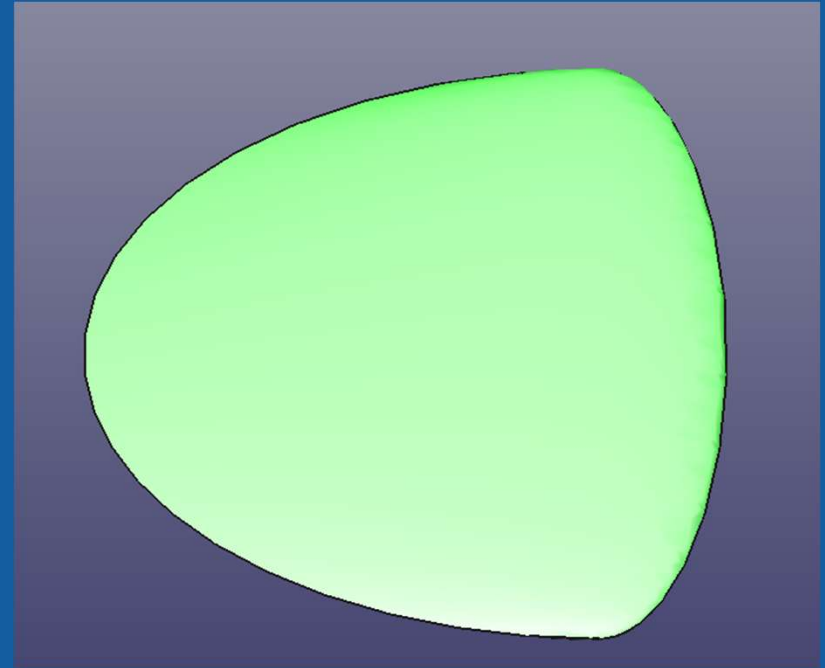
Simulations in ANSYS Fluent



CAD MODEL PREVIEW

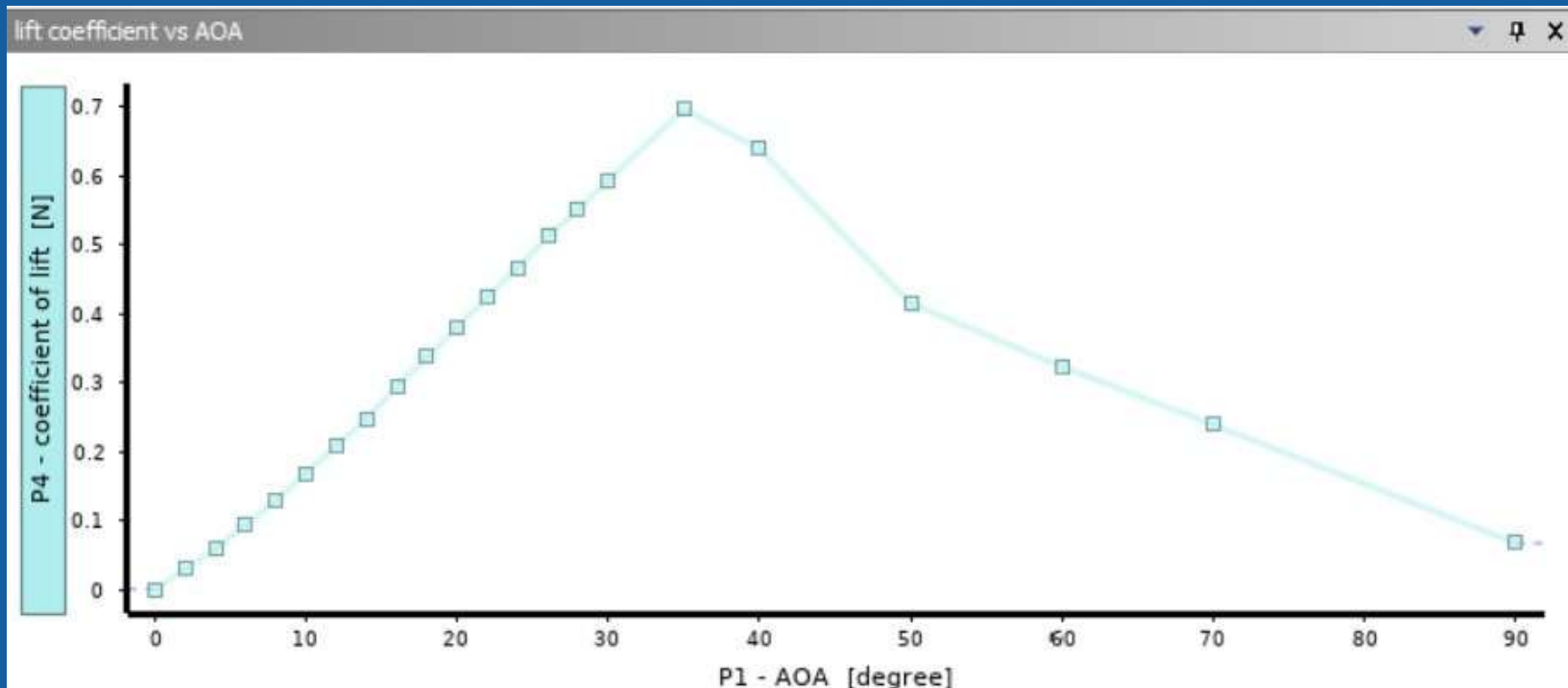


Side View



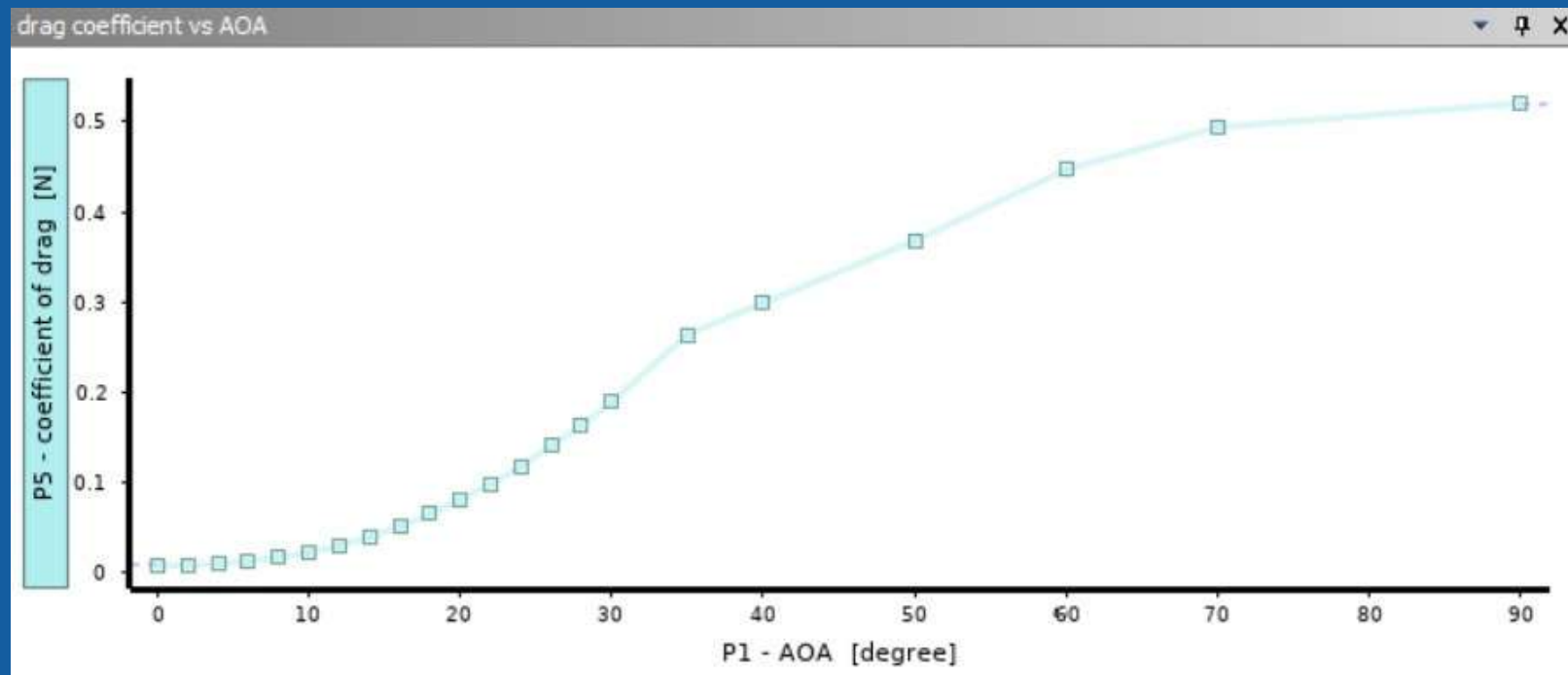
Planform View

ANSYS SIMULATION PERFORMANCE



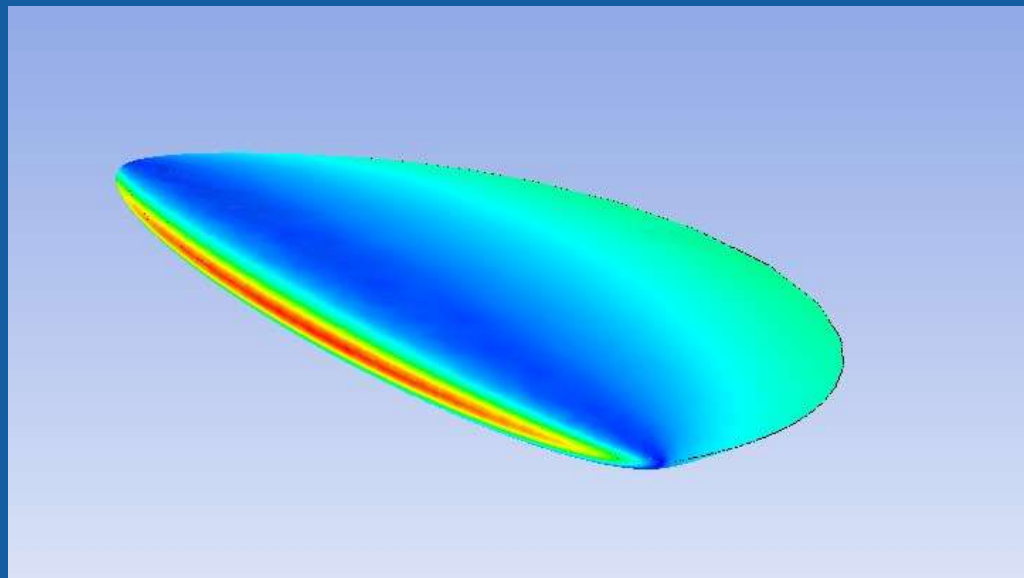
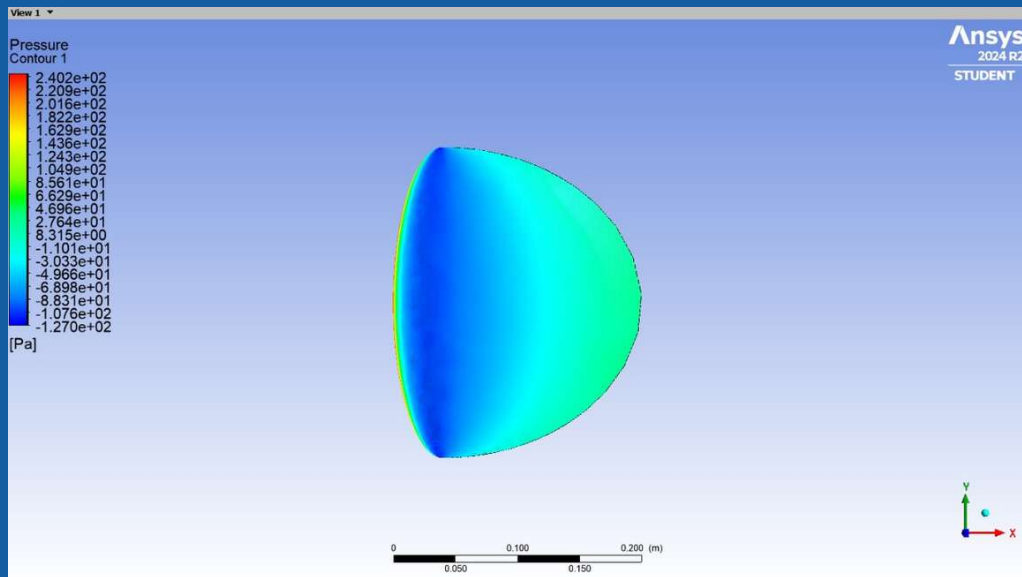
C_L vs Angle of Attack

ANSYS SIMULATION PERFORMANCE

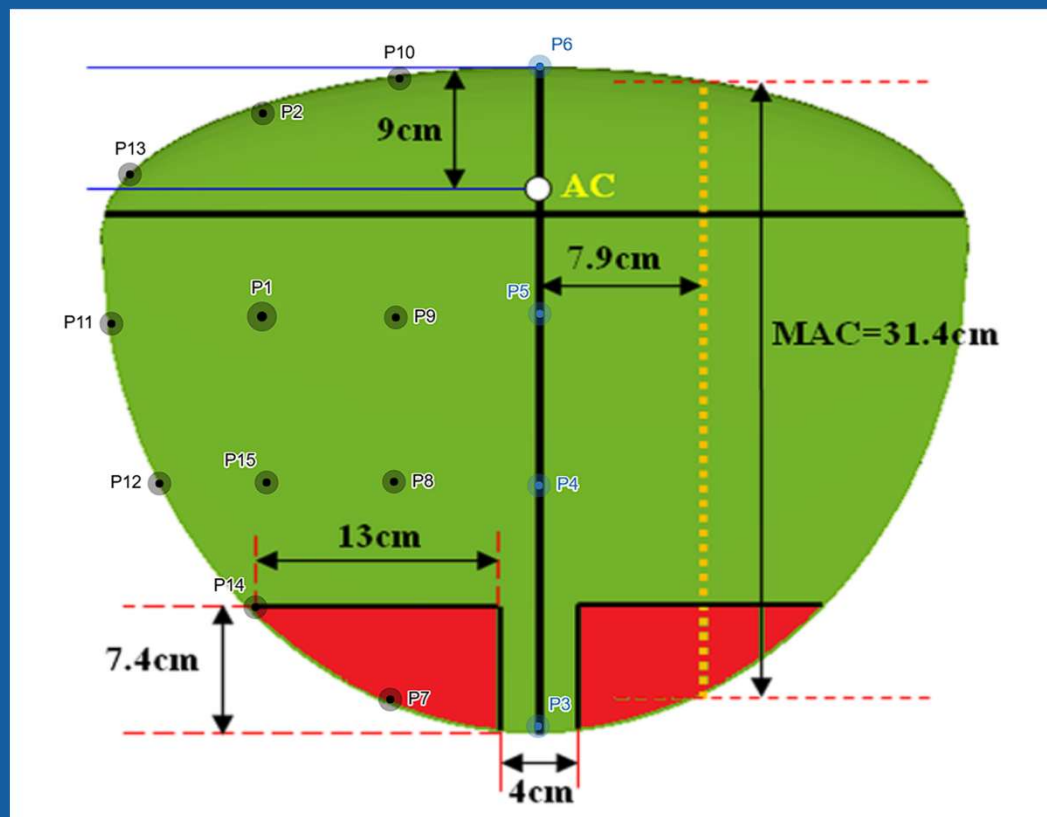


C_D vs Angle of Attack

ANSYS SIMULATION PERFORMANCE



PRESSURE PORTS ON CONTROL SURFACE



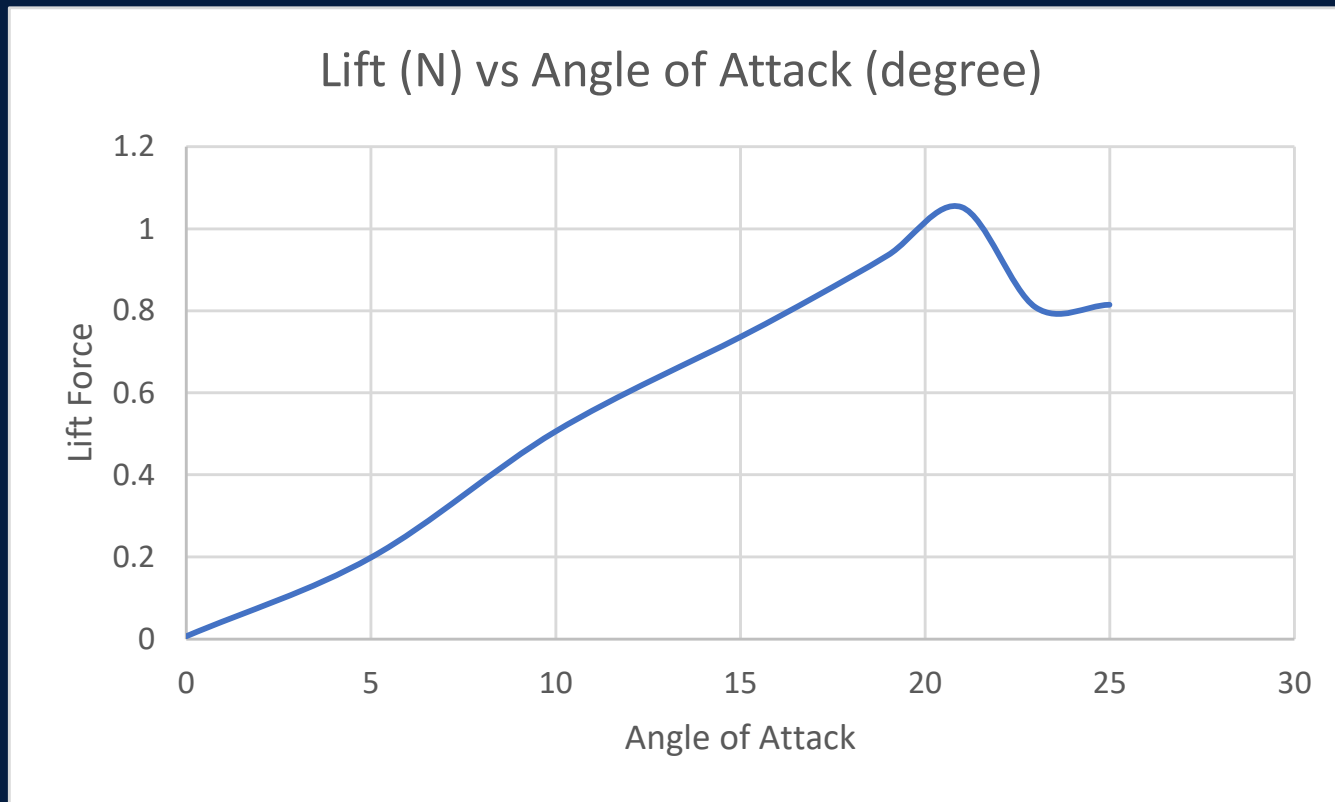
Force Measurements

AoA (degree)	Lift (N)
0	0.0068
5	0.19881
10	0.5061
15	0.7358
17	0.83228
19	0.9353
21	1.0519
23	0.80843
25	0.81474

AoA (degree)	Drag (N)
0	0.00412
5	0.02971
10	0.0871
15	0.14316
17	0.16875
19	0.20893
21	0.26144
23	0.42177
25	0.43597

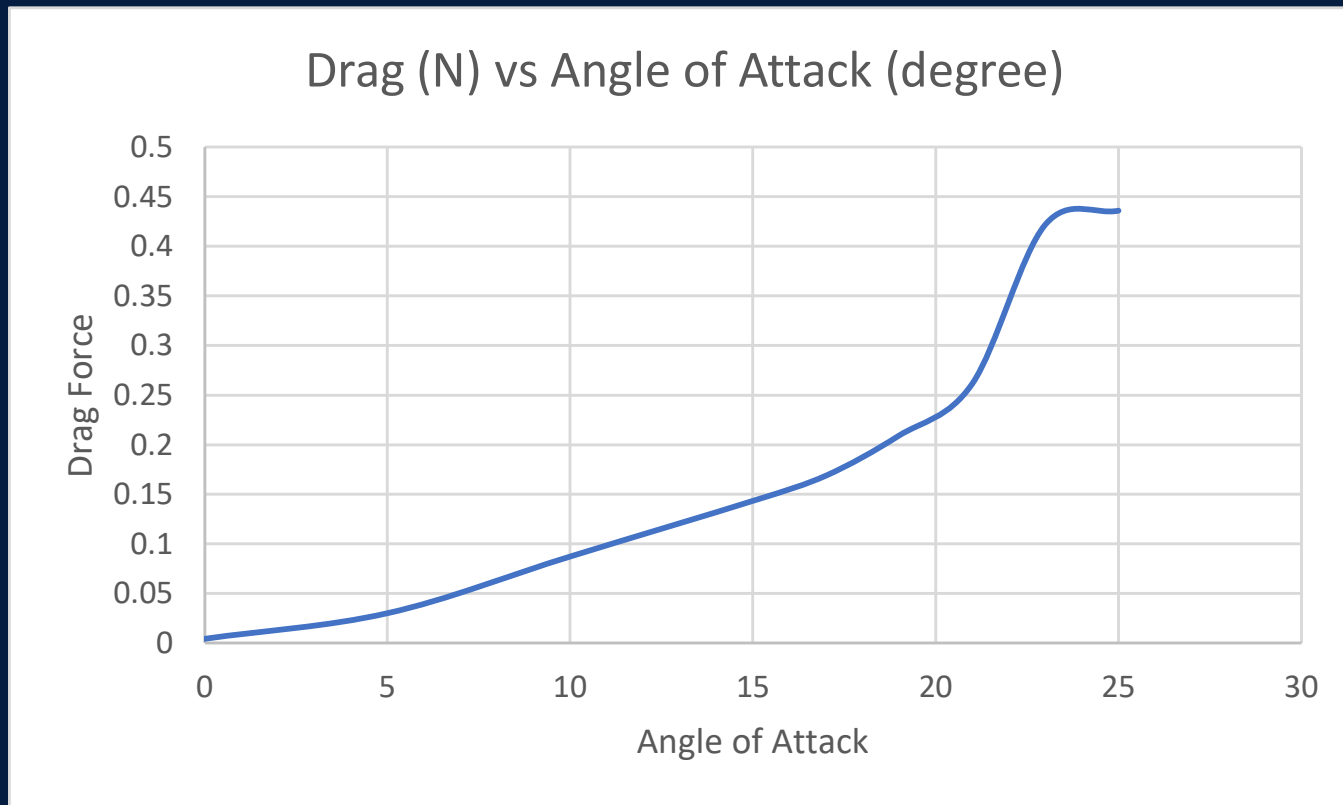
Table of Observations

Force Measurements



Plot: Lift (N) vs Angle of Attack (degree)

Force Measurements



Plot: Drag(N) vs Angle of Attack (degree)

Conclusion

Throughout the course of designing and analyzing Vought v-173, we studied its historical background ,read the background of development of the Flying Pancake. We went through its planform sketches and airfoils used also why it has very low take-off distance of 60m [close to VTOL]. We designed the CAD model and used ANSYS Fluent to create air-volume around it and get a feel about it lift and drag coefficients .Then we printed the model and proceeded to do force measurement in the wind tunnel to plot lift and drag forces vs angle of attack .From the experiment we observed the stall angle of attack to be 22 degrees which is the same of the actual value tested from it original model.



The trends observed in both experimental and simulation results confirmed the general aerodynamic behavior of the V-173 model. Overall, this project provided valuable insights into the design and testing process, highlighting the importance of both computational and experimental approaches in aerodynamic analysis. Future work could focus on refining the experimental model and exploring higher-resolution simulations to reduce discrepancies further and make it ready for flight testing.

This project underscores the utility of wind tunnel testing as a validation tool for simulations, offering a well-rounded approach for assessing the aerodynamic performance of unconventional aircraft designs.

THANK YOU